SAFEGUARDING FORECAST AI • DEPLOYMENT & STATISTICAL ECONOMETRICS AUDIT

Vercel Production Release & Feature Bloat vs. Sample Size Assessment

Date: 12 November 2026 **Branch / Commit:** `main` @ `92510fb`

Production Alias: https://safeguarding-forecast-ai.vercel.app

1. Vercel Production Deployment Summary

Production Alias	https://safeguarding-forecast-ai.vercel.app	**Status**	READY (HTTP 200)
Deployment URL	`https://safeguarding-forecast-23xdjvk4i-daniel-taggs-projects.vercel.app`	**Target**	`production`
Deployment ID	`dpl_99WQ6CuKtmsXXmX1nJFgnTLV3eZq`	**Build Duration**	`35s` (2,019 Vite modules)
Observability	Clean (`0` build/edge errors). Static Vite SPA served via Vercel Edge CDN with deterministic client-side snapshot fallback (`frontend/src/lib/api.ts` & `esotericTelemetrySnapshot.json`); no server-side log drains configured.		

2. Criterion Under Audit

> **Feature Bloat vs. Sample Size:** *On weekly academic series (≈ 190 teaching days or 39 active weeks per year), stacking all 13 frameworks unregularized will degrade out-of-sample error via variance inflation. The empirical literature consistently shows these features work best when paired with strict causal windowing, group-sparse pruning, and walk-forward ablation against a strong LightGBM or Negative Binomial baseline.*

3. Executive Verdict

- **Core Operational Forecasting (`backend/app/services/forecast_service.py`) — 0% Susceptible:** Completely decoupled from `esoteric_math`; baseline weights and out-of-sample predictions are verified bit-for-bit via IEEE-754 SHA-256 digests (`FrozenBaselineAdapter`).
- **Default Live Telemetry & Retrospective Stack (`STAGED_CORE_4` in `residual_stacker.py`) — Very Low Susceptibility:** Restricts candidates to 4 high-evidence modules (`f01, f02, f03, f07`), locks baselines as a fixed log-offset (`init_score = log(μ_base)`), applies block-permuted BorutaSHAP + Greedy 1-SE module quarantining, and enforces an exact baseline identity floor (`atol = 1e-12`) when 0 modules pass.
- **Opt-In 13-Module Gated Stack (`FULL_13_GATED` in `residual_stacker.py`) — Low–Moderate at Small N:** Quarantines all 13 modules cleanly at Folds `104` & `130`, but suffers a small-sample multiple-testing false positive (`f05` Ollivier-Ricci) at Fold `78` on `NB2` and `ZINB`, slightly degrading `CRPS` (`5.7751` → `5.8020`) and `Surge MAE` (`11.109` → `11.365`).
- **Legacy Direct-Refit Funnels (`esoteric_math/conditioning.py` + `ablation_runner.py`) — Moderate–High (Confirmed Variance Inflation):** Admitting up to 18 esoteric columns alongside 12 base predictors (`30` total features on N = 78..130 rows) and refitting baselines from scratch degrades departmental hierarchical `WRMSSE` across all three families (`+6.0%` on `NB2` , `+3.2%` on `ZINB` , and `+17.1%` on `TweedieLightGBM`).

4. Dimensionality vs. Effective Sample Size (p vs. N_{eff})

Across all 13 frameworks (`f01` – `f13` in `esoteric_math/features/__init__.py`), `EsotericFeatureSuite` emits **~135–145 candidate esoteric columns** on top of **12 domain baseline covariates**: `f01` Dual-Clock Fourier & Hilbert (~20), `f02` Haar/DB4 MODWT & Morlet (~21), `f03` Hawkes Contagion (~12), `f04` Takens TDA (9), `f05` Graph GFT & Ricci (7), `f06` Koopman & Tropical (7), `f07` Depth-3 Log-Signatures & Lévy Area (~17), `f08` Cellular Sheaf (8), `f09` p-adic & Poincaré (6), `f10` Optimal Transport (7), `f11` RMT Marchenko-Pastur (6), `f12` Fisher-Rao NB Geometry (8), and `f13` Fractional Diff & SOC (7).

On the 3-academic-year weekly benchmark (`156` calendar weeks, expanding origins at weeks `78` , `104` , and `130` with `20` holiday/closure/low-activity weeks per 52-week year), Fold 1 (`origin = 78`) provides only N **= 78 total rows** and N_{eff} **≈ 48–58 active teaching weeks**—an overparameterized $p \approx 3\,N_{eff}$ regime where unregularized stacking inevitably inflates out-of-sample variance.

5. Empirical Walk-Forward Scoreboard Evidence

5.1 Variance Inflation in Joint Refitting (`artifacts/esoteric_ablation_scoreboard.json`)

Although single-module walk-forward ablations confirm genuine standalone signal (e.g., `+f06` Koopman/Tropical lowers `NB2` `RMSE` from `12.552` → `10.879` and `CRPS` from `5.9991` → `5.3910` ; `+f01` Hilbert lowers `TweedieLightGBM` `CRPS` to `5.7523`), jointly refitting all 13 modules through `conditioning.py` inflates hierarchical `WRMSSE` and `WSPL` across every baseline:

Contender (`ablation_runner.py` , Folds 78 / 104 / 130)	WRMSSE	RMSSE	RMSLE	RMSE	Surge MAE	NB Dev	WSPL	CRPS
`control_nb2_glm` (Locked Baseline)	**0.5046**	**0.4573**	**0.6022**	12.552	12.968	2.3618	**0.1014**	5.9991
`esoteric_nb2_glm` (Joint 13-Module Funnel)	0.5351	0.4584	0.6102	**12.541**	**12.937**	**2.3296**	0.1081	**5.9881**
`control_zinb` (Locked Baseline)	**0.4951**	**0.4517**	**0.6010**	**12.390**	**12.659**	**2.3480**	**0.0992**	**6.0687**
`esoteric_zinb` (Joint 13-Module Funnel)	0.5109	0.4642	0.6092	12.692	13.256	2.3551	0.1055	6.2197
`control_tweedie_lgbm` (Locked Baseline)	**0.4601**	**0.3957**	0.8613	**10.905**	**11.869**	3.1496	**0.1031**	5.9768
`esoteric_tweedie_lgbm` (Joint 13-Module Funnel)	0.5389	0.4307	**0.6246**	11.803	12.455	**2.4313**	0.1225	**5.8832**

5.2 Variance Suppression in Post-Hoc Residual Quarantine (`artifacts/retrospective_5way_scoreboard.json`)

Freezing baselines and applying `PostHocResidualStacker` eliminates joint-refit degradation, while contrasting `STAGED_CORE_4` (zero false positives) against `FULL_13_GATED` (minor Fold-78 `f05` slippage):

Model Family & Rollout Arm	Admitted (F78 / F104 / F130)	WRMSSE	RMSLE	RMSE	Surge MAE	NB Dev	CRPS	PICP 95%
NB2 GLM — `1_frozen_baseline`	—	**0.4888**	0.5662	12.787	**11.109**	**1.8404**	**5.7751**	**94.87%**
NB2 GLM — `3_staged_core_4_stack`	`[]` / `[]` / `[]`	**0.4888**	0.5662	12.787	**11.109**	**1.8404**	**5.7751**	**94.87%**
NB2 GLM — `4_full_13_gated_stack`	`[f05]` / `[]` / `[]`	**0.4888**	**0.5609**	**12.784**	11.365	1.8510	5.8020	92.31%
ZINB — `1_frozen_baseline`	—	**0.5089**	0.5664	12.616	**10.396**	**1.8442**	**5.7304**	**93.59%**
ZINB — `3_staged_core_4_stack`	`[]` / `[]` / `[]`	**0.5089**	0.5664	12.616	**10.396**	**1.8442**	**5.7304**	**93.59%**
ZINB — `4_full_13_gated_stack`	`[f05]` / `[]` / `[]`	**0.5089**	**0.5627**	**12.603**	10.614	1.8545	5.7509	**93.59%**
Tweedie LGBM — `1_frozen_baseline`	—	**0.5225**	0.8286	12.879	12.485	3.1617	6.8536	75.64%
Tweedie LGBM — `3_staged_core_4_stack`	`[f03]` / `[]` / `[]`	0.5235	**0.7272**	**12.508**	**12.265**	**2.5745**	**6.4195**	**78.21%**
Tweedie LGBM — `4_full_13_gated_stack`	`[f03]` / `[]` / `[]`	0.5235	**0.7272**	**12.508**	**12.265**	**2.5745**	**6.4195**	**78.21%**

6. Evaluation Against Literature Safeguards

1. **Strict Causal Windowing (Strong — Verified to 1e-12):** `AdaptiveSchemaBridge` enforces strict $s < t$ lag shifting in forecast mode; `VintageSnapshotFilter` blocks late disclosures (`logged_date > t`); and `AcademicYearTopologyManager` resets Tier C cohort graphs (`f05, f08, f09`) every September 1 with a 4-week exponentially decayed warmup prior (`w_prior == 0` from Week 5 onward). Verified under +10^6 future-spike and backdated-disclosure suites.
2. **Group-Sparse Pruning (Strong in residual_stacker.py / Weak in conditioning.py):** `PostHocResidualStacker` pairs 25-trial 4-period circular block-permuted `BlockBorutaSHAPGate` (`≥ 18/25` shadow wins) with module-group `GreedyOneSEModuleSelector` (`1.15 × SE` gate), capping GLM offset heads at p_{max} = min(6, $\lfloor N_{eff} / 20 \rfloor$) with `VIF ≤ 5.0` and locking ZINB π_{zero}, while capping GBDT residual heads at `max_depth=2, num_leaves=4, reg_alpha=2.0, reg_lambda=10.0` .
3. **Walk-Forward Ablation Against Strong Baselines (Strong):** Evaluated across multi-origin expanding splits (`[78, 104, 130]` , `h = 26w`) at both college-wide and curriculum-area levels against exposure-offset `NB2GovernanceGLM` , `ZINBSparseRegimeModel` , and `TweedieLightGBMModel` .

7. Four Residual Code-Level Vulnerabilities & Recommended Hardening

1. Small-Sample Multiple-Testing Slippage in FULL_13_GATED (`residual_stacker.py` , lines 398–488)
On Fold `78` , `GreedyOneSEModuleSelector` builds K_{inner} = 3 splits with `val_len = 12` calendar weeks (~8–9 active weeks). With df = 2, `se_d` has high downward variance and `all(diffs > 0)` occurs at $(1/2)^3$ = 12.5% by chance per candidate module, letting `f05` slip past at N = 78. **Fix:** Scale the 1-SE multiplier by module multiplicity ($1.15\sqrt{(1 + \ln M_{cand})}$) or require $N_{eff} \geq 100$ before unlocking `FULL_13_GATED` .

2. Weekly N_{eff} Misses Holiday / Low-Activity Closures (`residual_stacker.py` , lines 91–112)
`compute_effective_temporal_sample_size` filters non-instructional rows only when `"is_instructional_day"` is present. Weekly panels (`cw_df` in `backend/app/features.py`) encode 20 closure/holiday weeks per year via `is_low_activity_week == 1` , causing weekly runs to treat N_{eff} = 52/yr instead of 32–39/yr and inflating p_{max} by ~35%. **Fix:** Discount rows with `is_low_activity_week > 0` when computing N_{eff}.

3. Un-Gated In-Sample Dispersion Fit in FisherRaoResidualCalibrator (`residual_stacker.py` , lines 806–880)
Unlike `GBDTInitScoreResidualHead` (which deactivates if a 20% chronological holdout fails to improve NB deviance), `FisherRaoResidualCalibrator` fits 12 depth-2 natural-gradient trees on up to 8 columns across 100% of training rows without an out-of-time validation gate—explaining the `94.87%` → `92.31%` `PICP_95` narrowing on Fold `78` . **Fix:** Gate calibrator activation on inner-holdout NB log-likelihood / CRPS gain.

4. In-Sample μ_{base} Residual Optimism & Legacy conditioning.py Caps (`residual_stacker.py` , lines 995–999; `conditioning.py` , lines 279–401)
`PostHocResidualStacker.fit` computes training residuals from in-sample `frozen_adapter.predict(train_df)` rather than out-of-fold baseline predictions (compressing GBDT residuals), while legacy `GBDTConditioningFunnel` permits `max_esoteric_cols = 18` with unblocked row permutation (`delta ≥ −1e−4`). **Fix:** Use out-of-fold baseline margins for non-parametric GBDT stacking and route `conditioning.py` callers through `PostHocResidualStacker` .